CODEFORCES
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Codeforces Round 1133 (Div. 1, Div. 2) is coming soon! Register here.

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

D. The Seven Bridges of Königsberg

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given an **undirected unweighted connected** graph with ***N*** nodes and ***M*** edges. There can be self loop or multiple edges. Your task is to determine whether an Eulerian Path exists in the graph.

In graph theory, an Eulerian path (also called an Eulerian trail or Eulerian walk) is a path in a graph that visits every edge exactly once and may start and end at different vertices. However, a vertex can be visited multiple times.

Input

The first line contains two integers N and M $(1 \leq N \leq 2 \times 10^5, 0 \leq M \leq 3 \times 10^5)$ — the number of vertices and the total number of edges.

The second line contains M integers $u_1, u_2, u_3 \ldots u_m$ $(1 \leq u_i \leq N)$ — where the i-th integer represents the node that is one endpoint of the i-th edge.

The third line contains M integers $v_1, v_2, v_3 \ldots v_m$ $(1 \leq v_i \leq N)$ — where the i-th integer represents the node that is other endpoint of the i-th edge.

The i'th edge of this graph is between the i'th node in the second line and the i'th node in the third line.

Output

If an Eulerian Path exists, print YES. Otherwise, print NO.

Examples

input Copy

```
5 10
5 5 5 2 2 2 3 3 4 2
2 3 1 3 4 1 4 1 2 4
```

output Copy

```
YES
```

input Copy

```
5 4
1 4 3 2
4 3 2 5
```

output Copy

```
YES
```

input Copy

```
8 7
4 4 6 6 3 1 8
6 5 3 2 7 8 7
```

output Copy

CSE221 Assignment 04 Fall 2026
Contest is running
8 days
Contestant

→ Submit?

Language: Java 21 64bit

Choose file: Choose File No file chosen

Success!

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Submit

→ Last submissions

Submission	Time	Verdict
397701036	Nov/07/2026 11:43	Accepted

```
NO
```

input Copy

```
7 6
3 5 7 6 4 2
5 7 6 4 2 1
```

output Copy

```
YES
```

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